

Clinical Risk Assessment Report: Patient 53

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 47.5% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 2.15x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT), which elevated the patient's absolute risk by +5.5%. Biologically, this feature reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 1.17 [Top 6%] (+5.5%)**

Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.19 [Avg (59th %ile)] (+4.1%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.31 [Bottom 22%] (+3.9%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.76 [Avg (75th %ile)] (+3.8%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Patient Age (Age) **Value: 55.0 [Bottom 22%] (+2.5%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: 1.43 [Top 3%] (+2.4%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.5 [Bottom 10%] (+2.3%)**

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: 1.16 [Top 9%] (+2.1%)**

Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

